

A Solution to $e^{\eta^2} = 2\eta^2 + 1$ and its relation to $\sqrt{2\pi/5}$

TM

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Abstract

We explore in great depth the unusually high accuracy (99.9918%) of the approximation, $\sqrt{2\pi/5}$, to Oseen's constant, 1.1209..., which has evaded a resolution for over a year.

1 Introduction

A solution to the Navier-Stokes equation in cylindrical coordinates, $(r, \theta, z) \in \mathbb{R}^3$, with the velocity field, $\vec{u} = \langle 0, u_\theta, 0 \rangle$, is the Lamb-Oseen vortex.

$$u_\theta(r, t) = \frac{\Gamma_0}{2\pi r} \left(1 - e^{-\frac{r^2}{4\nu t}} \right) \quad , \quad R(t) := \sqrt{4\nu t} \quad (1)$$

where Γ_0 is the circulation and ν is the kinematic viscosity. The Lamb-Oseen constant to 62 decimal places,

$$L_O := 1.12090642277853403197676673569063335993379778247221119947592997 \dots$$

defines a coefficient governing the vortex's core radius, where the global maximum and minimum of u_θ reside. We denote the core's radius as r_c , and simplify to get Equation (2).

$$\begin{aligned} 0 &= \left. \frac{\partial u_\theta}{\partial r} \right|_{r=r_c} \\ 0 &= \frac{\Gamma_0}{2\pi} \left(-\frac{1}{r^2} + \frac{1}{r^2} e^{-\frac{r^2}{R^2}} + \frac{2}{R^2} e^{-\frac{r^2}{R^2}} \right) \\ 0 &= -\frac{1}{r^2} + \frac{1}{r^2} e^{-\frac{r^2}{R^2}} + \frac{2}{R^2} e^{-\frac{r^2}{R^2}} \\ -\frac{1}{r^2} + \frac{1}{r^2} e^{-\frac{r^2}{R^2}} &= -\frac{2}{R^2} e^{-\frac{r^2}{R^2}} \\ \frac{1}{r^2} \left(1 - e^{-\frac{r^2}{R^2}} \right) &= \frac{2}{R^2} e^{-\frac{r^2}{R^2}} \\ e^{\frac{r^2}{R^2}} \left(1 - e^{-\frac{r^2}{R^2}} \right) &= 2 \frac{r^2}{R^2} \\ e^{\frac{r^2}{R^2}} - 1 &= 2 \frac{r^2}{R^2} \\ e^{\frac{r^2}{R^2}} &= 2 \frac{r^2}{R^2} + 1 \\ \text{Let, } \eta^2 &= \frac{r^2}{R^2} \\ \therefore e^{\eta^2} &= 2\eta^2 + 1 \end{aligned} \quad (2)$$

2 The Lambert-W method:

The first attempt yields the exact solution to Equation (2). We can solve as follows:

$$\begin{aligned}
e^{\eta^2} &= 2\eta^2 + 1 \\
e^{\frac{1}{2}} e^{-\frac{1}{2}} e^{\eta^2} &= 2 \left(\eta^2 + \frac{1}{2} \right) \\
\frac{1}{2} e^{-\frac{1}{2}} &= \left(\eta^2 + \frac{1}{2} \right) e^{-\eta^2 - \frac{1}{2}} \\
-\frac{1}{2} e^{-\frac{1}{2}} &= - \left(\eta^2 + \frac{1}{2} \right) e^{-\eta^2 - \frac{1}{2}} \\
W_{-1} \left(-\frac{1}{2} e^{-\frac{1}{2}} \right) &= W_{-1} \left(- \left(\eta^2 + \frac{1}{2} \right) e^{-\eta^2 - \frac{1}{2}} \right) \\
W_{-1} \left(-\frac{1}{2} e^{-\frac{1}{2}} \right) &= -\eta^2 - \frac{1}{2} \\
\eta &= \sqrt{-\frac{1}{2} - W_{-1} \left(-\frac{1}{2} e^{-\frac{1}{2}} \right)} \\
\frac{r}{R} &= \sqrt{-\frac{1}{2} - W_{-1} \left(-\frac{1}{2} e^{-\frac{1}{2}} \right)} \\
r &= R(t) \sqrt{-\frac{1}{2} - W_{-1} \left(-\frac{1}{2} e^{-\frac{1}{2}} \right)} \\
\therefore r_c &= (1.120906 \dots) \sqrt{4\nu t}
\end{aligned}$$

3 The Sinusoidal Approximation:

The second attempt is slightly more laborious. One may construct a function, $f : \mathbb{R} \rightarrow \mathbb{R}$, for Equation (2) as,

$$f(\eta) = 2\eta^2 + 1 - e^{\eta^2} \quad (3)$$

By the identity, $e^x := \sinh x + \cosh x$,

$$f(\eta) \equiv 2\eta^2 + 1 - \sinh(\eta^2) - \cosh(\eta^2) \quad (4)$$

we can make two observations with the substitution $x = \eta^2$:

1. $1 - \cosh x \sim 1$ iff $x \ll 1$
2. $2x - \sinh x \sim \sin x$ iff $x \ll 1$

The first observation is made by comparing the series expansion for $\cosh x$ to 1 for $x \ll 1$.

$$\begin{aligned}
& 1 - \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} \sim 1 \\
\iff & 1 - 1 + \sum_{n=1}^{\infty} \frac{x^{2n}}{(2n)!} \sim 1 \\
\iff & \frac{x^2}{2} + \frac{x^4}{4!} + O(x^6) \sim 1
\end{aligned}$$

The second observation involves comparing the two series for $\sin x$ and $\sinh x$ for $x \ll 1$:

$$\begin{aligned}
& 2x - \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} \sim \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} \\
\iff & 2x - \left\{ x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots \right\} \sim \left\{ x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right\} \\
\iff & - \sum_{n=1}^{\infty} \frac{x^{(4n+1)}}{(4n+1)!} \sim \sum_{n=1}^{\infty} \frac{x^{(4n+1)}}{(4n+1)!}
\end{aligned}$$

Therefore, one can deduce that there exists constants, $C, \delta \in \mathbb{R}^+$, such that

$$f(L_O) = C \sin(\delta L_O^2)$$

and hence,

$$f(\eta) \approx C \sin(\delta \eta^2) \tag{5}$$

Temporarily digressing, the amplitude, C , can be solved by taking the derivative of Equation (3) and setting it equal to zero:

$$\begin{aligned}
& 0 = f'(\eta) \\
\iff & 0 = 2\eta \left(2 - e^{\eta^2} \right) \\
\iff & \eta = 0, \sqrt{\ln 2} \\
\therefore & C = f(\sqrt{\ln 2})
\end{aligned}$$

Using the Newton-Raphson method, it is clear that an initial choice of η_k cannot be either 0 or $\sqrt{\ln 2}$.

$$\begin{aligned}
& \eta_{k+1} = \eta_k - \frac{f(\eta_k)}{f'(\eta_k)} \\
\iff & \eta_{k+1} = \eta_k - \frac{2\eta_k^2 + 1 - e^{\eta_k^2}}{2\eta_k (2 - e^{\eta_k^2})} \quad , \quad \eta_k \neq (0 \wedge \sqrt{\ln 2})
\end{aligned}$$

Performing this algorithm, an asymptote at $\eta_k = \sqrt{\ln 2}$ and a rapid convergence of $\eta_k \rightarrow 0$ for any $\eta_k \in (0, \ln 2)$ demonstrates that an initial value of $\eta_k > \sqrt{\ln 2}$ must be taken. Taking the roots of the approximation (5), we note that,

$$0 = C \sin(\delta \eta^2) = -C \sin(\delta_1 \eta^2 \pm \pi) = C \sin(\delta_2 \eta^2 \pm 2\pi)$$

where we denote δ_1 and δ_2 as the upper and lower bounds of δ . In the first case, the roots are found by the substitution $\eta = \sqrt{\ln 2}$:

$$\begin{aligned}
0 &= -C \sin(\delta_1 \eta^2 - \pi) \\
\iff \sin^{-1}(0) &= \delta_1 \eta^2 - \pi \\
\iff \pi &= \delta_1 \eta^2 \\
\iff \eta &= \sqrt{\frac{\pi}{\delta_1}}
\end{aligned} \tag{6}$$

and in the second case,

$$\begin{aligned}
0 &= C \sin(\delta_2 \eta^2 - 2\pi) \\
\iff \sin^{-1}(0) &= \delta_2 \eta^2 - 2\pi \\
\iff 2\pi &= \delta_2 \eta^2 \\
\iff \eta &= \sqrt{\frac{2\pi}{\delta_2}}
\end{aligned} \tag{7}$$

Finding the critical points of Equation (5), we have a set of values that δ cannot equal.

$$\begin{aligned}
0 &= \frac{d}{d\eta} [-C \sin(\delta \eta^2)] \quad , \quad \left\{ \eta = \sqrt{\ln 2} \right\} \Rightarrow \left\{ \sin(\eta^2) \neq 0 \right\} \\
0 &\neq 2\delta \ln 2 \cos(\delta \ln 2) \\
\cos^{-1}(0) &\neq \delta \ln 2 \\
\delta &\neq \left\{ \frac{\pi}{2 \ln 2}, \frac{3\pi}{2 \ln 2}, \frac{5\pi}{2 \ln 2}, \dots \right\}
\end{aligned}$$

Thus, we have the following two cases:

Case 1: Lower bound on δ for $-C \sin(\delta_1 \eta^2 \pm \pi)$.

Setting up the inequality and substituting into Equation (6), we have a lower bound on δ_1 .

$$\begin{aligned}
&\sqrt{\frac{\pi}{\delta_1}} > \sqrt{\ln 2} \\
\text{where, } \left\{ \sqrt{\frac{\pi}{\frac{3\pi}{2 \ln 2}}}, \sqrt{\frac{\pi}{\frac{5\pi}{2 \ln 2}}}, \dots \right\} &\not> \sqrt{\ln 2} \\
&\text{leaving, } \sqrt{\frac{\pi}{\frac{\pi}{2 \ln 2}}} > \sqrt{\ln 2} \\
&\therefore \delta_1 > \frac{\pi}{2 \ln 2}
\end{aligned}$$

Case 2: Upper bound on δ for $C \sin(\delta_2 \eta^2 \pm 2\pi)$.

Likewise, substituting invalid values of δ into Equation (7),

$$\begin{aligned} & \sqrt{\frac{2\pi}{\delta_2}} > \sqrt{\ln 2} \\ \text{where, } & \left\{ \sqrt{\frac{2\pi}{\frac{5\pi}{2\ln 2}}}, \sqrt{\frac{2\pi}{\frac{7\pi}{2\ln 2}}}, \dots \right\} \not> \sqrt{\ln 2} \\ \text{leaving, } & \left\{ \sqrt{\frac{2\pi}{\frac{\pi}{2\ln 2}}}, \sqrt{\frac{2\pi}{\frac{3\pi}{2\ln 2}}} \right\} > \sqrt{\ln 2} \\ & \therefore \text{choose } \delta_2 < \frac{3\pi}{2\ln 2} \end{aligned}$$

Between each case, we conclude that if $\eta = \sqrt{\frac{\pi}{\delta_1}}$, then $\delta_1 > \frac{\pi}{2\ln 2}$, and if $\eta = \sqrt{\frac{2\pi}{\delta_2}}$, then $\delta_2 < \frac{3\pi}{2\ln 2}$. Therefore, if $\eta = \sqrt{\frac{2\pi}{\delta}}$, then

$$\sqrt{\frac{4\ln 2}{3}} < \eta < \sqrt{\ln 4} \iff \frac{\pi}{\ln 2} < \delta < \frac{3\pi}{2\ln 2} \quad (8)$$

If we take the average between $\sqrt{\ln(2^{4/3})}$ and $\sqrt{\ln(4)}$, η 's approximation becomes $\bar{\eta} = \sqrt{\ln(2^{5/3})} \approx 1.0748$; only 95.889% accurate. Because $\sqrt{\ln 4} \approx 1.1774$ is closer to 1.1209, and because $\delta > \frac{\pi}{\ln 2}$, if,

$$\delta = \left\lceil \frac{\pi}{\ln 2} \right\rceil \equiv 5$$

then,

$$\eta = \sqrt{\frac{2\pi}{\left\lceil \frac{\pi}{\ln 2} \right\rceil}} = \sqrt{\frac{2\pi}{5}} < \sqrt{\frac{2\pi}{\frac{\pi}{\ln 2}}}$$

This produces a 99.99% accuracy to L_O .

4 Conclusions

By approximating Equation (3) in terms of the sine function, determining the roots, and finding inequalities for $\delta \in \eta$ in each case, the mystery of $\sqrt{2\pi/5}$ is uncovered by rounding the lower value of δ that yields an η that is closer to $\sqrt{\ln 4}$ than $\sqrt{\ln(2^{4/3})}$.